OpenShift Troubleshooting Scenarios – Top 30 Scenarios Every OpenShift Expert Should Know

Introduction

Red Hat OpenShift, built on Kubernetes, provides a robust container orchestration platform for deploying and managing enterprise applications. However, troubleshooting OpenShift environments can be complex due to the multiple layers involved — nodes, pods, networking, storage, authentication, and cluster configuration.

This document highlights **30 essential troubleshooting scenarios** that every OpenShift expert should know. These cases reflect common real-world issues in operations, deployments, and performance management, along with methods to identify root causes and resolve them efficiently.

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Cluster and Node-Level Troubleshooting

Scenario 1: Node Not Ready

Symptom: A node shows "NotReady" in oc get nodes.
Cause: Kubelet or network connectivity issues.
Resolution:

- Check node status: oc describe node <node-name>
- Restart kubelet service: systemctl restart kubelet
- Verify SDN (OpenShift SDN or OVN-Kubernetes) is active.

Scenario 2: Cluster Operators Degraded

Symptom: Operators show "Degraded" or "Unavailable" in oc get co.
Cause: Failed component updates or unreachable API endpoints.
Resolution:

- Run: oc get co -A
- Review operator logs: oc logs -n openshift-cluster-version deploy/cluster-version-operator
- Identify specific failing operator and restart the component.

Scenario 3: ETCD Performance Degradation

Symptom: Cluster performance slow; control plane commands timeout.
Cause: ETCD disk latency or memory exhaustion.
Resolution:

- Check ETCD health: oc exec -n openshift-etcd etcd-<pod> -- etcdctl endpoint health
- Free disk space and monitor IOPS.
- Ensure low-latency SSD-backed storage.

Scenario 4: Node Disk Pressure

Symptom: Pods evicted due to disk pressure.
Cause: Log accumulation or container images occupying disk space.
Resolution:

- Clean unused images: crictl rmi --prune
- Rotate logs regularly.
- Check /var/lib/containers/storage for excessive data.

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Scenario 5: Node Certificate Expiration

Symptom: Node fails to join after reboot.
Cause: Expired kubelet certificate.
Resolution:

- Check certificate: oc get csr
- Regenerate certificate: oc adm certificate approve <csr-name>

Pod and Application-Level Troubleshooting

Scenario 6: Pod CrashLoopBackOff

Symptom: Application pod repeatedly restarts.
Cause: Misconfiguration, missing environment variables, or readiness probe failures.
Resolution:

- Check pod logs: oc logs <pod>
- Verify configuration secrets and probes.
- Fix startup scripts or resource requests.

Scenario 7: ImagePullBackOff

Symptom: Image not pulled from registry.
Cause: Invalid credentials or unavailable image.
Resolution:

- Verify registry credentials: oc get secret
- Ensure image exists in repository.
- Redeploy with correct image tag.

Scenario 8: Pending Pods

Symptom: Pods stuck in Pending state.
Cause: Insufficient resources or missing tolerations.
Resolution:

- Describe pod: oc describe pod <pod>
- Check node capacity and taints.

- Adjust resource requests or use node selectors.

Scenario 9: Init Container Failures

Symptom: Init container fails before main pod starts.
Cause: Misconfigured init script or network dependency failure.
Resolution:

- View logs of init container.
- Fix script paths or dependency order.

Scenario 10: Application Deployment Fails

Symptom: oc rollout status shows failure.
Cause: Missing configuration or invalid environment.
Resolution:

- Use: oc describe deployment <name>
- Review readiness/liveness probes and configuration maps.

Networking Troubleshooting

Scenario 11: Service Not Accessible

Symptom: Internal or external traffic cannot reach a service.
Cause: Incorrect service type or network policy restrictions.
Resolution:

- Check service endpoints: oc get endpoints <svc>
- Validate NetworkPolicies.
- Ensure routes and DNS entries exist.

Scenario 12: Route Not Working

Symptom: HTTP 503 errors from external route.
Cause: Ingress controller misconfiguration or TLS issues.
Resolution:

- Inspect route status: oc describe route <name>

- Verify ingress controller pod health.
- Recreate TLS certificate if expired.

Scenario 13: DNS Resolution Failure

Symptom: Pods can't resolve service names.
Cause: CoreDNS failure.
Resolution:

- Check CoreDNS pods in openshift-dns namespace.
- Restart failed pods.

Scenario 14: NetworkPolicy Blocking Traffic

Symptom: Pods unable to communicate across namespaces.
Resolution:

- Review active policies: oc get networkpolicy -A
- Modify or delete restrictive policies.

Scenario 15: LoadBalancer Service Not Assigned

Symptom: No external IP for LoadBalancer service.
Cause: Cloud provider integration issue.
Resolution:

- Inspect controller logs.
- Ensure cloud credentials operator is configured properly.

Storage and Persistent Volume Issues

Scenario 16: PVC Stuck in Pending

Cause: No available storage class or dynamic provisioner.
Resolution:

- Verify: oc get storageclass
- Configure default storage class if missing.

Scenario 17: PV Not Bound

Cause: Mismatch between PV and PVC attributes.
Resolution:

- Ensure same access modes and storage sizes.
- Delete and recreate PVC.

Scenario 18: ReadOnly FileSystem in Pods

Cause: Underlying node disk or permission issue.
Resolution:

- Restart node if necessary.
- Check persistent volume health on backend storage.

Scenario 19: Data Not Persisting After Restart

Cause: PVC not mounted correctly.
Resolution:

- Verify mount paths in deployment YAML.
- Check for deleted or mismatched PVCs.

Scenario 20: GlusterFS/CephFS Connectivity Issues

Resolution:

- Validate storage daemon health.
- Check firewalls and network paths between nodes and storage cluster.

Authentication and Access Troubleshooting

Scenario 21: User Unable to Login

Cause: OAuth configuration or LDAP connectivity issue.
Resolution:

- Check `oc get oauth` configuration.
- Validate identity provider URLs.

Scenario 22: Unauthorized Errors After Login

Cause: RBAC misconfiguration.
Resolution:

- Review roles: `oc get rolebindings -A`
- Grant permissions using `oc adm policy add-role-to-user`.

Scenario 23: Token Expiration Issues

Resolution:

- Check API server token configuration.
- Increase token TTL via OAuth settings.

Scenario 24: ServiceAccount Missing

Cause: Incorrect namespace or name.
Resolution:

- Recreate missing service account.
- Bind proper roles to it.

Monitoring and Performance

Scenario 25: High API Server Latency

Cause: ETCD or network latency.
Resolution:

- Monitor metrics with Prometheus.
- Scale control plane nodes if necessary.

Scenario 26: Pod Resource Exhaustion

Symptom: CPU/Memory throttling or eviction.

Resolution:

- Adjust resource requests and limits.
- Use Horizontal Pod Autoscaler.

Scenario 27: Metrics Not Available

Cause: Prometheus or node exporter issues.
Resolution:

- Restart monitoring pods.
- Check `openshift-monitoring` namespace health.

Scenario 28: Logging Stack Down

Symptom: Logs not visible in Kibana or Loki.
Resolution:

- Check Elasticsearch or Loki pods.
- Ensure sufficient disk space.

Upgrade and Configuration Troubleshooting

Scenario 29: Cluster Upgrade Fails

Cause: Incompatible operator versions or failing pre-checks.
Resolution:

- Review: `oc get clusterversion`
- Fix failing components before retrying upgrade.

Scenario 30: Configuration Drift After Update

Cause: Manual edits overwritten by operator reconciliation.
Resolution:

- Avoid manual changes to managed resources.
- Use Custom Resource Definitions (CRDs) properly.

Conclusion

OpenShift troubleshooting requires a deep understanding of its layered architecture — from container runtime and networking to security and storage. By mastering these **30 core troubleshooting scenarios**, OpenShift experts can efficiently identify problems, maintain cluster stability, and ensure reliable application performance.
Continuous monitoring, proper configuration management, and adherence to Red Hat best practices are key to preventing these issues in production environments.

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